



Omii Chauhan

Roll No.: 23MMB0A31

Bachelor of Technology — Metallurgy and Materials Science

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Education

National Institute of Technology, Warangal

2023 – 2027

Bachelor of Technology — Metallurgy and Materials Science

Telangana State Board of Intermediate Education

2022

Class XII (Higher Secondary) — 96%

Central Board of Secondary Education

2020

Class X (Secondary School) — 93%

Experience

ADEN (Y Combinator)

Feb 2026 – Present

Core Framework Contributor

Remote

- **Concurrency & Resilience:** Engineered a per-key asynchronous locking strategy to resolve a critical TOCTOU race condition, and refactored the file-spillover and session checkpointing pipelines to isolate corrupted states, preventing silent execution failures on resume.
- **Performance Engineering:** Refactored legacy synchronous tool integrations (Exa, Apollo) to native asynchronous execution using `httpx.AsyncClient`, eliminating event-loop starvation.
- **Advanced Swarm Consensus:** Implemented the core engine for the Neutrosophic Scoring Module (RFC #7179), introducing multi-dimensional confidence logic for swarm decision-making.
- **Security & Protocol Hardening:** Patched critical directory traversal vulnerabilities across 13 routing endpoints and engineered a custom DNSSEC validation pipeline bypassing system resolvers to enforce Authenticated Data (AD) verification.

Rajputana Vehicles Private Limited

May 2025 – June 2025

Data Analyst Intern

Madhya Pradesh, India

- Developed an automated ETL data pipeline using Python (Pandas) to process raw business data, eliminating 10+ hours/week of manual reporting.
- Architected customer segmentation models utilizing K-Means clustering on 50k+ transaction records, improving campaign targeting precision by 15%.
- Designed interactive KPI dashboards in Power BI to track EV adoption rates and inventory turnover, enabling data-backed strategic decisions for senior management.

Technical Projects

FinTech AI: Credit Risk Prediction Engine | [GitHub](#) | [Live Demo](#)

2024 – 2025

Real-time loan default prediction using ensemble ML models

- **Stack:** Python, XGBoost, Random Forest, Scikit-learn, Docker, FastAPI
- Developed an end-to-end ML pipeline (Random Forest & XGBoost ensemble) to predict borrower default risk, encompassing robust feature engineering and probability calibration.
- Engineered a containerized batch processing engine (Docker) capable of scoring 10,000+ loan applications asynchronously via FastAPI workers, computing Exposure at Default (EAD) and Loss Given Default (LGD) metrics in under 5 seconds.

AI Multi-Strategy Fund Platform | [GitHub](#)

2025

Autonomous agent-driven investment lifecycle platform

- **Stack:** Python, LangGraph, FastAPI, Next.js 14, Recursive Risk Parity, Redis
- Built a multi-agent orchestration workflow using LangGraph and Redis queues, where independent agents generated allocation signals and portfolio summaries asynchronously.
- Implemented hierarchical risk parity-based allocation logic to actively rebalance portfolios using historical market data.

Visual Workflow Builder | [GitHub](#) | [Live Demo](#)

2026

Production-grade node-based pipeline editor with real-time DAG validation

- **Stack:** React, React Flow, Zustand, FastAPI (Python), REST APIs, CORS
- Architected an extensible **BaseNode** component abstraction, reducing UI boilerplate by 40% and allowing rapid deployment of 5+ custom node types (LLM, Database, File, Timer, Email).
- Engineered dynamic variable parsing with custom regex to dynamically render input handles on text nodes, backed by a **Zustand** global state machine synchronization pipeline.
- Developed a FastAPI backend using **Kahn's Algorithm (Topological Sort)** to perform real-time Directed Acyclic Graph (DAG) validation, cycle detection, and pipeline analysis in $O(V + E)$ complexity.

Technical Skills

Languages: Python (asyncio), SQL, C++, TypeScript

Data & ML: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, Feature Engineering

Backend & Infrastructure: FastAPI, PostgreSQL, Snowflake, Redis, Docker, AWS, DNSSEC

Frontend & Tools: React, Next.js, Git, CI/CD, DevOps

Leadership & Achievements

Leadership: Head of PR & Sponsorship, Technozion & Springspree (Managed 50+ students & corporate sponsors).

Certifications & Simulations: HackerRank SQL (Advanced), J.P. Morgan Quantitative Research, EA Software Engineering.